

9-2c The Effects of a Tariff

The Isolandian economists next consider the effects of a

tariff (a tax on goods produced abroad and sold domestically)—a tax on imported goods.

The economists quickly realize that a tariff on textiles will have no effect if Island becomes a textile exporter. If no one in Island is interested in importing textiles, a tax on textile imports is irrelevant. The tariff matters only if Island becomes a textile importer.

Concentrating their attention on this case, the economists compare welfare with and without the tariff.

Figure 4 shows the Isolandian market for textiles. Under free trade, the domestic price equals the world price. A tariff raises the price of imported textiles above the world price by the amount of the tariff. Domestic suppliers of textiles, who compete with foreign suppliers of imported textiles, can now sell their textiles for the world price plus the amount of the tariff. As a result, the price of textiles—both imported and domestic—rises by the amount of the tariff and is, therefore, closer to the price that would prevail without trade.

Figure 4

The Effects of a Tariff

A tariff, a tax on imports, reduces the quantity of imports and moves a market closer to the equilibrium that would exist without trade. Total surplus falls by an amount equal to area $D + F$. These two triangles represent the deadweight loss from the tariff.

The change in price affects the behavior of domestic buyers and sellers. Because the tariff raises the price of textiles, it reduces the domestic quantity demanded from Q_1^D to Q_2^D and raises the domestic quantity supplied from Q_1^S to Q_2^S . *Thus, the tariff reduces the quantity of imports and moves the domestic market closer to its equilibrium without trade.*

Now consider the gains and losses from the tariff. Because the tariff raises the domestic price, domestic sellers are better off, and domestic buyers are worse off. In addition, the government raises revenue, which it can use for public purposes. To measure these gains and losses, we look at the changes in consumer surplus, producer surplus, and government revenue, as we did in [Chapter 8](#). These changes are summarized in the table in [Figure 4](#).

Before the tariff, the domestic price equals the world price. Consumer surplus, the area between the demand curve and the world price, is area **A + B + C + D + E + F**. Producer surplus, the area between the supply curve and the world price, is area **G**. Government revenue equals zero. Total surplus, the sum of consumer surplus, producer surplus, and government revenue, is area **A + B + C + D + E + F + G**.

Once the government imposes a tariff, the domestic price exceeds the world price by the amount of the tariff. Consumer surplus is now area **A + B**. Producer surplus is area **C + G**. Government revenue, which is the size of the tariff multiplied by the quantity of after-tariff imports, is the area **E**. Thus, total surplus with the tariff is area **A + B + C + E + G**.

To determine the total welfare effects of the tariff, we add the change in consumer surplus (which is negative), the change in producer surplus (positive), and the change in

government revenue (positive). We find that total surplus in the market decreases by the area **D + F**. This fall in total surplus is the *deadweight loss* of the tariff.

A tariff causes a deadweight loss because a tariff is a type of tax. Like most taxes, it distorts incentives and pushes the allocation of scarce resources away from the optimum. In this case, we can identify two effects. First, when the tariff raises the domestic price of textiles above the world price, it encourages domestic producers to increase production from Q_1^S to Q_2^S . Even though the cost of making these incremental units exceeds the cost of buying them at the world price, the tariff makes it profitable for domestic producers to manufacture them nonetheless. Second, when the tariff raises the price that domestic textile consumers have to pay, it encourages them to reduce their consumption of textiles from Q_1^D to Q_2^D . Even though domestic consumers value these incremental units at more than the world price, the tariff induces them to cut back their purchases. Area D represents the deadweight loss from the overproduction of textiles, and area F represents the deadweight loss from the underconsumption of textiles. The total deadweight loss of the tariff is the sum of these two triangles.

FYI

Import Quotas: Another Way to Restrict Trade

Beyond tariffs, another way that nations sometimes restrict international trade is by putting limits on how much of a good can be imported. In this book, we will not analyze such a policy, other than to point out the conclusion: Import quotas are much like tariffs. Both tariffs and import quotas reduce the quantity of imports, raise the domestic price of the good, decrease the welfare of domestic consumers, increase the welfare of domestic producers, and cause deadweight losses.

There is only one difference between these two types of trade restriction: A tariff raises revenue for the government, whereas an import quota generates surplus for those who obtain the permits to import. The profit for the holder of an import permit is the difference between the domestic price (at which she sells the imported good) and the world price (at which she buys it).

Tariffs and import quotas are even more similar if the government charges a fee for the import permits. Suppose the government sets the permit fee equal to the difference between the domestic price and the world price. In this case, the entire profit of permit holders is paid to the government in permit fees, and the import quota works exactly like a tariff. Consumer surplus, producer surplus, and government revenue are precisely the same under the two policies.

In practice, however, countries that restrict trade with import quotas rarely do so by selling the import permits. For example, the U.S. government has at times pressured Japan to “voluntarily” limit the sale of Japanese cars in the United States. In this case, the Japanese government allocates the import permits to Japanese firms, and the surplus from these permits accrues to those firms. From the

standpoint of U.S. welfare, this kind of import quota is worse than a U.S. tariff on imported cars. Both a tariff and an import quota raise prices, restrict trade, and cause deadweight losses, but at least the tariff generates revenue for the U.S. government rather than profit for foreign producers.

Chapter 9: Application: International Trade: 9-2c The Effects of a Tariff

Book Title: WGU Economics

Printed By: Mike Benton (mbento6@wgu.edu)

© 2021 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may be reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.